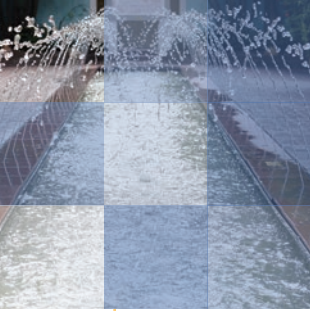




The Polynomial Remainder Theorem

You can use the polynomial remainder theorem to evaluate a polynomial function.

Polynomial Remainder Theorem

If a polynomial $p(x)$ is divided by $x - a$, then the remainder is $p(a)$.

$$\begin{array}{ccccccc} p(x) & = & q(x) & \cdot & (x-a) & + & p(a) \\ \uparrow & & \uparrow & & \uparrow & & \uparrow \\ \text{dividend} & = & \text{quotient} & \cdot & \text{divisor} & + & \text{remainder} \end{array}$$

To prove this theorem, recall that you can use multiplication to check a quotient.

Let $p(x)$ represent the dividend, $q(x)$ represent the quotient, and $r(x)$ represent the remainder. If $x - a$ is the divisor, then the formula can be written as follows:

$$p(x) = q(x) \cdot (x - a) + r(x)$$

Now substitute a for x .

$$p(a) = q(a) \cdot (a - a) + r(a) \quad \text{Substitute } a \text{ for } x.$$

$$p(a) = q(a) \cdot 0 + r(a) \quad a - a = 0$$

$$p(a) = r(a) \quad \text{Zero Property of Multiplication}$$

Therefore, the remainder is the value of the polynomial evaluated for a .

Evaluating a Polynomial

Sometimes synthetic division is an easier and quicker method to evaluate a polynomial, especially when a calculator is not readily available.

EXAMPLE 1

Given $f(x) = x^4 - 3x^3 + x^2 - 1$, find $f(-2)$.

SOLUTION

Use synthetic division and the remainder theorem.

$$\begin{array}{r|rrrrr} -2 & 1 & -3 & 1 & 0 & -1 \\ & & -2 & 10 & -22 & 44 \\ \hline & 1 & -5 & 11 & -22 & 43 \end{array}$$

The remainder is 43, so $f(-2) = 43$.

Substitute -2 for x in $f(x)$ to check.

$$f(-2) = (-2)^4 - 3 \cdot (-2)^3 + (-2)^2 - 1 = 16 + 24 + 4 - 1 = 43 \quad \blacksquare$$

Using the Remainder Theorem to Graph a Polynomial

EXAMPLE 2

Graph the function by using the remainder theorem.

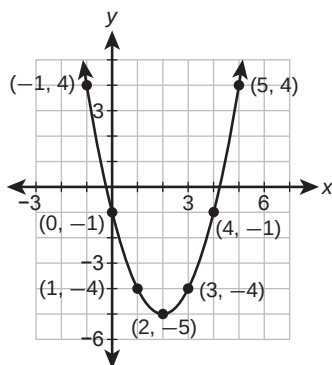
A $f(x) = x^2 - 4x - 1$

SOLUTION

Use synthetic division repeatedly to find and plot ordered pairs.

Organize your work in a table. List the coefficients of the dividend at the top. Perform the steps mentally and write only the numbers that appear in the bottom row. Plot each pair $(a, f(a))$.

	1	-4	-1
-1	1	-5	4
0	1	-4	-1
1	1	-3	-4
2	1	-2	-5
3	1	-1	-4
4	1	0	-1
5	1	1	4

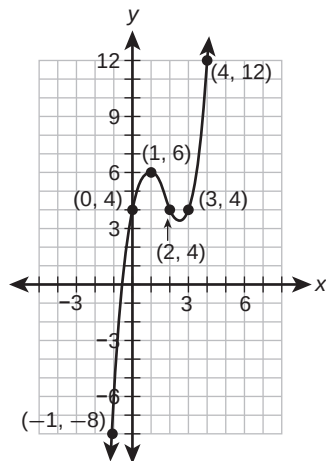


B $g(x) = x^3 - 5x^2 + 6x + 4$

SOLUTION

Use synthetic division repeatedly to find and plot ordered pairs.

	1	-5	6	4
-1	1	-6	12	-8
0	1	-5	6	4
1	1	-4	2	6
2	1	-3	0	4
3	1	-2	0	4
4	1	-1	2	12



Application: Projectile Motion

EXAMPLE 3

A baseball is hit into the air with an initial vertical velocity of 20 m/s. The height of the baseball after t seconds is modeled by the polynomial function $h(t) = -9.8t^2 + 20t + 1$, where $h(t)$ is the height in meters (neglecting air resistance). What is the height of the baseball after 1.5 s?

SOLUTION

The height of the baseball in meters after t seconds is $h(t)$. To find the height of the baseball after 1.5 s, find $h(1.5)$.

Method 1 Use the remainder theorem.

Divide $-9.8t^2 + 20t + 1$ by $t - 1.5$, using synthetic division.

$$\begin{array}{r|rrrr} 1.5 & -9.8 & 20 & 1 & \\ & & -14.7 & 7.95 & \\ \hline & -9.8 & 5.3 & 8.95 & \end{array}$$

The remainder is 8.95, so $h(1.5) = 8.95$.

► **Think About It** You can use one method to solve the problem and the other method to check your answer.

Method 2 Substitute 1.5 for t in $h(t)$.

$$\begin{aligned} h(t) &= -9.8t^2 + 20t + 1 \\ h(1.5) &= -9.8 \cdot 1.5^2 + 20 \cdot 1.5 + 1 \\ &= -9.8 \cdot 2.25 + 20 \cdot 1.5 + 1 \\ &= 8.95 \end{aligned}$$

The height of the baseball after 1.5 s is 8.95 m. ■